

NATIONAL JUNIOR COLLEGE
PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

SUBJECT
CLASS

REGISTRATION
NUMBER

CHEMISTRY

Paper 3 Free response

9746/03

Mon 14 Sep 2009

Cover page

2 hours

Answer any four questions.

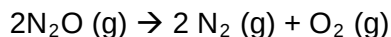
Please indicate the questions you have attempted by putting a “√” against the question number.

	Put “√” against the question attempted	Examiner’s use
Question 1		/ 20
Question 2		/ 20
Question 3		/ 20
Question 4		/ 20
Question 5		/ 20
		Subtotal: /80

[Turn over

Answer any **four** questions.

- 1 Nitrous oxide**, commonly known as **happy gas** or **laughing gas**, due to the euphoric effects of inhaling it. This property has led to its recreational use as an inhalant drug. Nitrous oxide is rather unstable and will readily decompose into its elements as shown.



- (a) Define electronegativity and compare the relative electronegativity of the nitrogen and oxygen atoms.
- (b) (i) Draw the dot-and-cross diagram for N_2 . Hence, suggest a dot-and-cross diagram for N_2O , stating its shape and bond angle.
- (ii) Real gases deviate from ideality by a different extent. Which of the following three gases, N_2O , N_2 or O_2 , would require the highest temperature to behave like an ideal gas? Explain your answer. [9]
- (c) The rate of decomposition of nitrous oxide may be followed by measuring the partial pressure of N_2O .

Time/s	Partial pressure of N_2O /kPa
0	25.0
1000	20.0
2500	15.0
4500	10.0
6000	7.0
7500	5.0

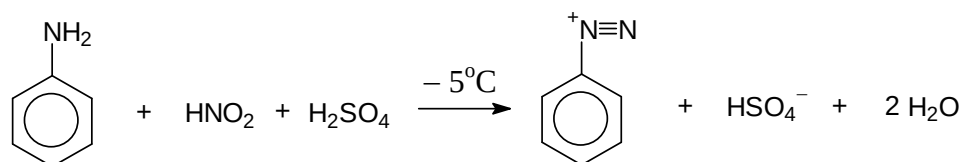
- (i) Determine the order of reaction with respect to N_2O by making use of the above data.
- (ii) Calculate the value of the rate constant, k . [3]

- 1 (d) Nitrogen and oxygen can react at high temperatures in the car exhaust pipe to form the oxides of nitrogen, one of which is nitrogen monoxide, NO.

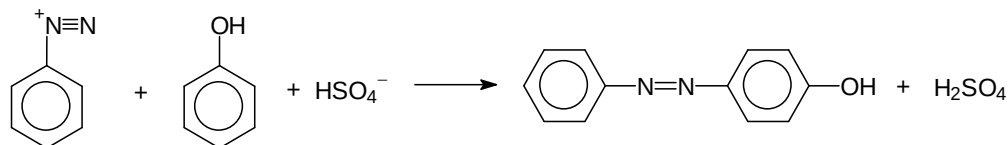
NO reacts with oxygen and water to form nitrous acid, HNO_2 .

Nitrous acid is very useful in organic synthesis and can be used to synthesise brightly coloured compounds called *azodyes*. The following shows the steps in the preparation of azodye from nitrous acid, phenylamine and phenol in the presence of sulfuric acid.

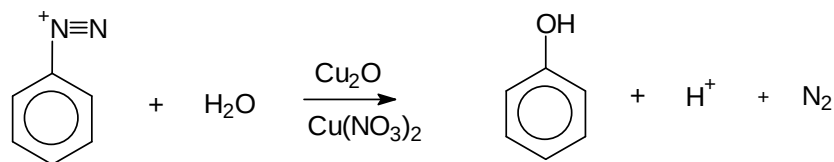
Step 1



Step 2



- (i) State the type of reaction phenol has undergone in step 2.
- (ii) Step 2 only occurs with phenol but not with benzene even though they undergo the same type of reaction. Suggest and explain why it is so. [3]
- (e) The intermediate diazonium ion in step 1 can also be used to synthesize phenol.



- (i) Give the structural formula of the compound that might be produced when diazonium salt were to react with ethanol rather than water. Suggest why the reaction in (e) is favourable. [2]
- (ii) Using phenol as a starting reactant, suggest a two-step synthesis to form the following compound, 4-hydroxyphenylamine. [3]



[Total: 20]

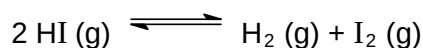
[Turn over

- 2 (a) AgNO_3 was added to a mixture containing $0.001 \text{ mol dm}^{-3}$ each of sodium iodide and sodium iodate.
(Given: K_{sp} of Silver iodide, AgI , 8.52×10^{-17} , K_{sp} of silver iodate(V), AgIO_3 , 3.17×10^{-8} .)

- (i) Calculate the minimum concentration of silver ions required to cause precipitation of the following
(I) Silver iodide
(II) Silver iodate
- (ii) Hence calculate the concentration of iodide ions in the solution when the first trace of silver iodate appears.

[4]

- (b) Sodium iodide can react with concentrated phosphoric(V) acid to give hydrogen iodide gas.
Hydrogen iodide gas can decompose into hydrogen and iodine at 500 K according to the following equation:



0.0250 mol of HI was injected into a closed vessel and allowed to reach equilibrium at 500 K. The contents of the equilibrium mixture was titrated quickly with 0.1 mol dm^{-3} of aqueous sodium thiosulfate. It was found that 28.00 cm^3 of sodium thiosulfate was required for complete reaction with iodine.

- (i) Explain why the mixture must be titrated quickly.
- (ii) Calculate the equilibrium constant, K_c , for the thermal decomposition of hydrogen iodide.
- (c) Hydrogen iodide can also undergo organic reaction with alkenes to form halogenoalkane. Describe the mechanism for the reaction between hydrogen iodide and propene; and hence explain why one of the products is formed in greater percentage.

[4]

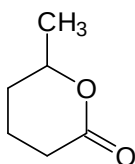
- (d) Ethanoyl iodide, CH_3COI and iodoethanoic acid, CH_2ICOOH are halogen containing organic compound **A** and **B** respectively. When both are added to separate portions of water, two solutions with pH values 0.5 and 3.0 are obtained.
Suggest, with explanations, which pH value is associated with each of **A** and **B**.

[3]

- 2 (e) An iodoalkane **C**, $C_6H_{11}I$, which has a plane of symmetry, was heated with ethanolic KOH under reflux and a compound **D**, C_6H_{10} , was obtained. Under strongly oxidising conditions, compound **D** gave only one organic product **E**, $C_6H_{10}O_3$.

Compound **E** gives yellow precipitate with alkaline iodine solution. Compound **E** also gives brisk effervescence with aqueous sodium carbonate.

When compound **E** was reduced with hydrogen gas and nickel, compound **F**, $C_6H_{12}O_3$ was obtained. When compound **F** was heated with concentrated sulfuric acid, a sweet smelling liquid with the following structure was isolated:



Deduce the structures of compounds **C** to **F**, with clear reasoning.

[6]

[Total: 20]

- 3 Ethanoic acid is an organic acid which gives vinegar its sour taste and pungent smell. 25.0 cm^3 of 1.0 mol dm^{-3} ethanoic acid has a pK_a value of 4.75.

(a) Calculate the pH of the solution.

[1]

2 separate experiments of acid-base neutralization were carried out to determine the enthalpy of neutralization. The concentrations of solutions used below are 1.0 mol dm^{-3}

Experiment 1		Experiment 2	
Volume of CH_3COOH / cm^3	25.0	Volume of HCl / cm^3	25.0
Volume of $NaOH$ / cm^3	30.0	Volume of $NaOH$ / cm^3	30.0
Temperature rise/ $^{\circ}C$	5.5	Temperature rise/ $^{\circ}C$	6.2

(b) (i) Explain why the temperature rise in **experiment 1** is smaller?

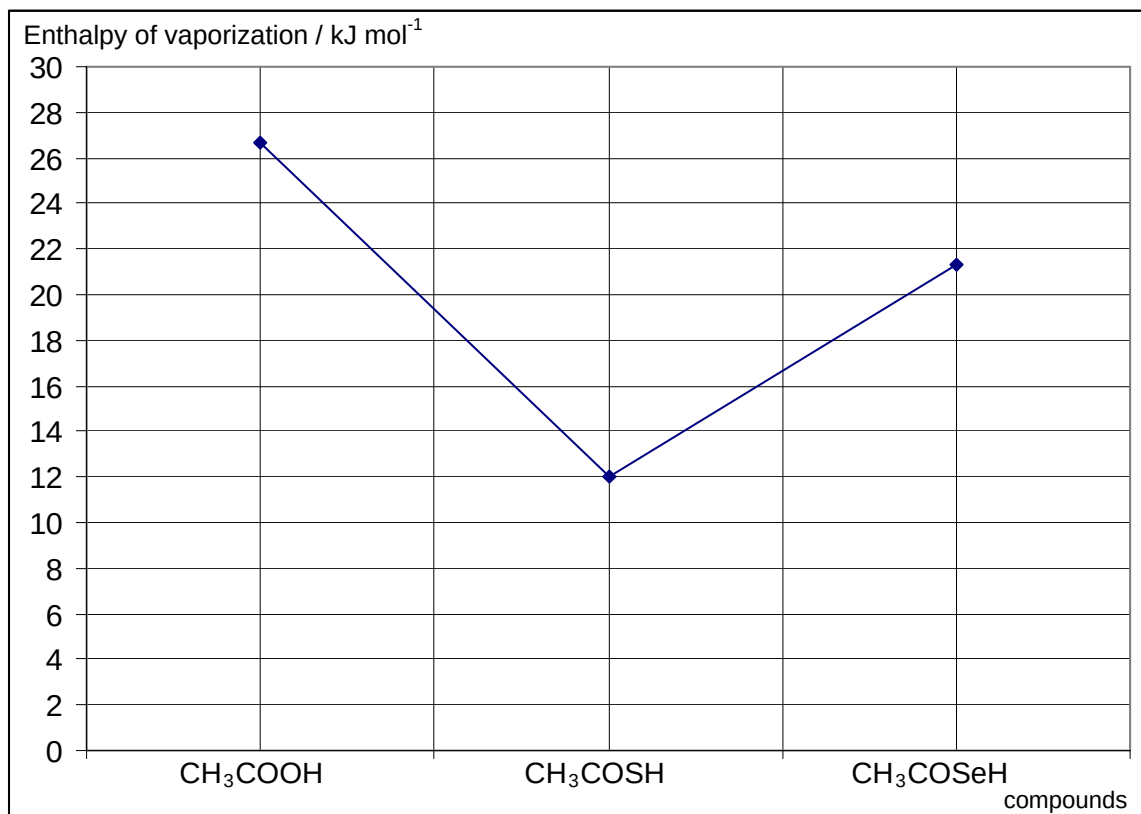
[2]

(ii) Determine the enthalpy of neutralization of **experiment 2** given that the specific heat capacity of water is $4.2\text{ Jg}^{-1}\text{K}^{-1}$.

[2]

[Turn over]

- 3 (c) The graph below shows the enthalpy of vaporization of CH_3COOH , CH_3COSH and CH_3COSeH .

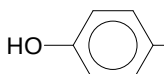


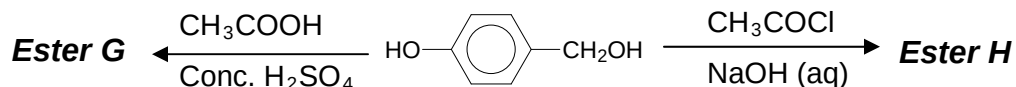
- (c) (i) Explain the above trend. [3]
- (ii) By making use of the above graph, deduce the enthalpy of formation of H-bonding. [2]
- (iii) Ethanoic acid, CH_3COOH , undergo dimerisation in vapour state.
- (I) Draw a displayed formula of the dimer formed.
- (II) Determine the enthalpy of dimerisation by referring to (c)(ii).
- (III) Given that the entropy of dimerisation is $-130 \text{ J mol}^{-1} \text{ K}^{-1}$, explain why dimerisation is spontaneous at low temperatures under the same pressure.

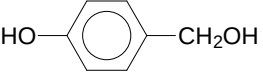
[5]

[Turn over]

3 (d)

Ethanoic acid can react with  to form two different esters under suitable conditions.



- (i) Draw the structural formulae of **ester G** and **H**.
- (ii) Briefly explain why an alkaline medium is necessary for the formation of **ester H**.
- (iii) Suggest a simple chemical test that can be used to distinguish ethanoic acid and .

[5]

[Total: 20]

4 Lautarite, also known as calcium iodate(V), $\text{Ca}(\text{IO}_3)_2$, is a commonly used chemical as a dough conditioner to strengthen the texture of bread dough.

- (a) Draw a dot-and-cross diagram for calcium iodate(V).

[2]

Iron can also form iodate(V) compounds but it is not used as a dough conditioner because it is coloured.

- (b) (i) Explain why calcium iodate(V) is white, while iron compounds are coloured.
- (ii) Iron metal has a much higher first ionization energy than calcium metal. Account for the observation.

[5]

At high temperatures, calcium iodate(V) undergo decomposition to form a white solid, a colourless gas and violet iodine vapour.

- (c) (i) Construct an equation, with state symbols, to represent the above decomposition process.
- (ii) State and explain the thermal stability trend of group II iodate(V).

[5]

[Turn over]

- 4 (d) The violet iodine vapour formed from the above decomposition process is bubbled into a closed vessel, containing a gaseous alkene. The violet vapour was decolourised and a liquid compound, with relative molecular mass of 310 was obtained.

- (i) Given that this alkene exhibit geometric isomerism, deduce its identity by explaining the observations made.
- (ii) Iodine can also react with iron metal.

Use $E_{\text{red}}^{\ominus}$ values from the *Data Booklet* to predict what will be formed when excess Fe(s) is added to iodine solution.

[5]

- (e) The reaction between iodate(V) and iodide ions is represented below:



The reaction is repeated in 3 separate experiments, using starch as an indicator.

	Time taken to form blue black colouration
Reaction of iodate(V) and iodide	1 day
Reaction of iodate(V) and iodide in the presence of Co^{2+}	10 min
Reaction of iodate(V) and iodide in the presence of Ca^{2+}	1 day

Account for the above observation by making specific reference to the role of the cations where applicable in the above reaction mixtures.

[3]

[Total: 20]

- 5 Chloroethane and chlorobenzene have been used widely in the industries. Chloroethane is a colourless gas at room temperature and is used as a refrigerant while chlorobenzene is a colourless liquid at room temperature and is used to make pesticides.

Chloroethane undergoes nucleophilic substitution with NH_3 readily to form ethylamine while chlorobenzene does not react with NH_3 .

- (a) Describe the mechanism of the reaction between chloroethane and NH_3 .

[2]

- (b) Explain:

- (i) the difference in the reactivities of $\text{CH}_3\text{CH}_2\text{Cl}$ and $\text{C}_6\text{H}_5\text{Cl}$ towards NH_3 .
- (ii) the different physical states of $\text{CH}_3\text{CH}_2\text{Cl}$ and $\text{C}_6\text{H}_5\text{Cl}$ at room temperature.

[4]

[Turn over]

- 5 (c) (i) The reaction between chloroethane and ammonia always run into the difficulty in obtaining primary amine as the pure product. Explain why it is so and hence suggest a condition to increase the yield of ethylamine.
- (ii) Chloroethane and chloromethane, present in 1:1 molar ratio, can both further react with ethylamine to form 3 types of tertiary amines.

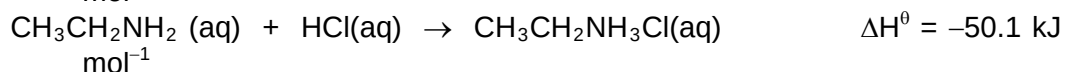
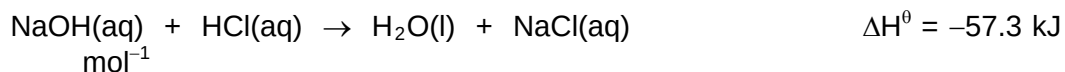
Draw the structures of these 3 types of tertiary amines and hence state the ratio in which they are formed.

[4]

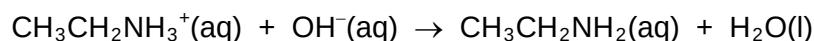
- (d) Ethylamine can form an acid derivative under suitable conditions. Starting with chloromethane, suggest a four-step synthesis to make $\text{CH}_3\text{CH}_2\text{NHCOCH}_3$, using $\text{CH}_3\text{CH}_2\text{NH}_2$ in one of the steps.

[3]

- (e) NaOH and $\text{CH}_3\text{CH}_2\text{NH}_2$ react with hydrochloric acid as shown in the equations below:



- (i) Using the information above, calculate the enthalpy change for the following reaction by drawing an energy cycle.



- (ii) Two separate samples of 25.0 cm^3 of 0.1 mol dm^{-3} aqueous sodium hydroxide and 25.0 cm^3 of 0.1 mol dm^{-3} aqueous ethylamine were titrated against 0.1 mol dm^{-3} hydrochloric acid.

(given: K_b of ethylamine = $2.5 \times 10^{-5} \text{ mol dm}^{-3}$)

Sketch their titration curves on the same axis, labeling the important points along the curve.

[7]

[Total: 20]

